

Topic 4

Geometry

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

Learning focus

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

Topic map

4.1	Geometrical language and angle facts
4.2	Polygons and symmetry
4.3	Congruence and similarity
4.4	Circle theorems
4.5	Constructions, loci and bearings

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

1. Angles and polygons

Angles on a straight line total 180° , around a point 360° and inside a triangle 180° . Parallel lines yield corresponding and alternate equal angles and co-interior supplementary angles. A polygon with n sides has interior angle sum $(n - 2) \times 180^\circ$.

Worked example

A regular octagon has interior angle $(8 - 2) \times 180^\circ / 8 = 135^\circ$.

Exam check: Name the angle rule in a reasoning question; a bare numerical value may not show why.

2. Congruence and similarity

Congruent triangles match exactly in side lengths and angles, often by SSS, SAS, ASA or RHS. Similar figures have equal corresponding angles and lengths in a common ratio k . Corresponding areas change by k^2 and volumes by k^3 . Match corresponding vertices before calculating.

Worked example

Similar triangles have sides 4 cm and 10 cm corresponding; $k = 2.5$. If small area is 12 cm^2 , larger area $= 12 \times 2.5^2 = 75 \text{ cm}^2$.

Exam check: A length scale factor must be squared for areas.

3. Circle geometry

An angle at the centre is twice the angle at the circumference on the same arc. Angles in the same segment are equal; the angle in a semicircle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° . The radius is perpendicular to the tangent at contact.

Worked example

If a centre angle subtending an arc is 116° , the corresponding circumference angle is 58° .

Exam check: Make sure the angles stand on the same arc; a sketch prevents using the wrong theorem.

4. Constructions and loci

A perpendicular bisector is the locus equidistant from two fixed points. An angle bisector is the locus equidistant from the sides of an angle. The locus at a fixed distance from a point is a circle; from a line it is two parallels plus semicircular end caps for a finite segment. Show arcs in a compass construction.

Worked example

Points 3 cm from A form a circle of radius 3 cm centred at A. Intersect it with a perpendicular bisector to satisfy both conditions.

Exam check: Read whether the question asks for a shaded region, a boundary or individual intersection points.

5. Bearings and transformations of shape

Bearings are measured clockwise from north and written with three digits. In a scale diagram, draw north lines at both points; reverse bearings differ by 180° modulo 360° . Reflections preserve size but reverse orientation; rotations preserve size and orientation.

Worked example

The bearing of B from A is 065° . The bearing of A from B is 245° .

Exam check: Always measure a bearing from north at the starting point of that journey.

Before the practice paper

Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.

Practice paper

Topic 4 · Geometry

Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

Candidate name:	
Class / date:	

Total: 28 marks

Section A · Multiple choice

Choose one answer for each question.

1. Sum of angles in a triangle? [1]

- A 90°
- B 180°
- C 270°
- D 360°

2. Interior angle sum of a hexagon? [1]

- A 360°
- B 540°
- C 720°
- D 900°

3. One exterior angle of a regular decagon? [1]

- A 18°
- B 36°
- C 72°
- D 144°

4. If lengths scale by 3, areas scale by [1]

- A 3
- B 6
- C 9
- D 27

5. A diameter subtends what angle at the circumference? [1]

- A 45°
- B 90°
- C 180°
- D 360°

6. The radius meets a tangent at [1]

- A 30°
- B 45°
- C 60°
- D 90°

7. The perpendicular bisector is equidistant from [1]

- A two fixed points
- B two intersecting lines
- C a line and a point
- D the sides of an angle

8. Reverse bearing of 045° ? [1]

- A 135°
- B 180°
- C 225°
- D 315°

Section B · Structured questions

9. A regular polygon has an exterior angle of 24° .

(a) Find its number of sides. [2]

(b) Find one interior angle. [1]

(c) Find the sum of its interior angles. [2]

[Total: 5]

10. Two similar triangles have corresponding sides of 6 cm and 15 cm. The smaller triangle has area 18 cm^2 .

(a) Calculate the length scale factor. [1]

(b) Calculate the area scale factor. [2]

(c) Find the larger triangle area.

[2]

[Total: 5]

11. In a circle, the centre angle AOB is 124° and C is on the major arc AB.

(a) Find angle ACB.

[2]

(b) State the theorem used.

[1]

(c) Find angle OAB in the isosceles triangle AOB.

[2]

[Total: 5]

12. The bearing of Q from P is 118° .

(a) Find the bearing of P from Q.

[2]

(b) Find the angle clockwise from north to a due west direction.

[1]

(c) Explain why a bearing must be measured at the point of departure.

[2]

[Total: 5]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.
Check each part independently.

Section A · Answers and reasons

1. B — 180° . All interior angles of any triangle total 180° .
2. C — 720° . $(6-2) \times 180^\circ = 720^\circ$.
3. B — 36° . $360^\circ/10 = 36^\circ$ for one exterior angle.
4. C — 9. Area changes by the square of the length factor: $3^2 = 9$.
5. B — 90° . The angle in a semicircle is a right angle.
6. D — 90° . The tangent is perpendicular to the radius at contact.
7. A — two fixed points. Every point on a perpendicular bisector is equidistant from the endpoints.
8. C — 225° . The reverse bearing is $045^\circ + 180^\circ = 225^\circ$.

Section B · Worked solutions

9. a) $360^\circ/24^\circ = 15$ sides [2]. b) $180^\circ - 24^\circ = 156^\circ$ [1]. c) $(15 - 2) \times 180^\circ = 2340^\circ$ [2].

10. a) $k = 15/6 = 2.5$ [1]. b) $k^2 = 6.25$ [2]. c) $18 \times 6.25 = 112.5 \text{ cm}^2$ [2].

11. a) Angle $ACB = 124^\circ/2 = 62^\circ$ [2]. b) Angle at centre is twice angle at circumference on the same arc [1]. c) $OA = OB$; each base angle $= (180^\circ - 124^\circ)/2 = 28^\circ$ [2].

12. a) $118^\circ + 180^\circ = 298^\circ$ [2]. b) 270° [1]. c) A bearing is the clockwise direction from north at its starting point; changing the starting point changes the reference north line [2].

Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,
www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf. Paper style informed by the June 2026 papers and examiner report provided by the user. All practice questions are original.